Detecting Anomalous Antenna Coverage Using Distance Rate

Juthamart Jaruensuk, Watid Phakphisut, Chuwong Phongcharoenpanich,

Thongchai Wijitpornchai, and Tanun Jaruvitayakovit

School of Engineering, King Mongkut's Institute of Technology Ladkrabang, Bangkok, Thailand, 10520
E-mail: 64011077@kmitl.ac.th, watid.ph@kmitl.ac.th, chuwong.ph@kmitl.ac.th, thongchw@ais.co.th, and tanunjar@ais.co.th

Abstract Anomalous antennas diminish the quality of service in mobile networks. Therefore, this research endeavors to identify the anomalous antenna where the coverage distance is not as expected. After the anomalous antennas are observed, the mobile network operator (MNO) can prepare the maintenance or replacement process precisely. The anomalous antenna is detected by comparing the estimated distances derived from antenna parameters such as mechanical tilt (*M tilt*), electrical tilt (*E tilt*), vertical beam width (*VBW*), and horizontal beam width (*HBW*) with distances inferred from Timing Advance (TA) values. By comparing both measured distances, we can identify the degraded antenna offering valuable insights of the coverage of mobile network, facilitating precise maintenance interventions and enhancing overall network reliability.

Keyword Anomalous antenna, Coverage distance, Timing Advance, Vertical Beam Width

1. Introduction

Mobile network systems serve as the backbone of modern economies and societies. Reliable and extensive communication is essential for the growth of businesses, education, healthcare, and public services. However, the quality of user experience depends on the efficiency of mobile networks, specifically, the installation and configuration of transmitters with antennas being a critical component.

Deteriorated or malfunctioning antennas can significantly degrade the performance of transmitted power, leading to weak signals, dropped calls, slow data connections, and a subpar user experience. Furthermore, faulty antennas can adversely affect the overall performance of mobile networks, resulting in increased energy consumption and reduced network capacity.

Consequently, the identification and monitoring of degraded antennas are crucial for maintaining the quality of services. Regular inspections enable mobile network operator (MNO) to address issues promptly and continuously improve network performance. Common antenna anomalies include short coverage ranges, incorrect orientation, and abnormal radiation patterns.

This research focuses on short coverage range case whereby the detect degraded antennas is detected by comparing both measured and estimated distances. The findings of degraded antennas are expected to facilitate the efficient automated monitoring systems, reducing operating costs and enhancing the quality of services.

2. Base Station Parameters

The coverage range of a base station can be estimated based on antenna characteristics such as

- Mechanical Tilt (*M tilt*): The physical angle which the antenna is inclined relative to the vertical. It is adjusted mechanically by tilting the entire antenna structure.

- Electrical Tilt (*E tilt*): The electrical adjustment of the antenna's radiation pattern. It allows fine-tuning of the vertical beam without physically moving the antenna.

- Vertical Beamwidth (*VBW*): The angular width of the antenna's main lobe in the vertical direction. It determines the vertical coverage area of the signal.

- Antenna Height: The physical height of the antenna above the ground. This affects the signal propagation path and can influence coverage, especially in areas with obstacles like buildings or trees.

- Digital Elevation Model (*DEM*): A digital representation of the Earth's surface, providing elevation data at regular intervals. In this study, we used *DEM* of both the base station and the grid where the MS is located.

- The approximate height of the mobile station (MS) assumes 1.5 meters, which is a typical height for a human holding a mobile device.

The relationship of parameters is illustrated in Figure 1.

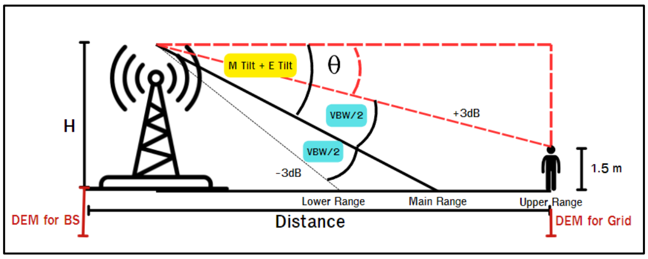

Fig.1. The relationship of parameters from antenna characteristics

3. Timing Advance

Timing Advance (*TA*) [4] is a parameter employed in mobile communication systems, such as GSM and LTE, to synchronize the transmission timing of mobile stations (MSs) and the base station (BS). The *TA* is used to compensate the propagation delay that occurs as the radio signal travels from the MS to the BS.

The measured transmission range of the antenna can be determined by analyzing the *TA* value. By comparing the measured *TA* with the calculated value derived from base station parameters, the antenna performance can be obtained.

The *TA* is typically expressed in units of bit periods or microseconds. In this work, *TA* must be converted to meters. This conversion requires considering the propagation speed of radio waves, which is approximately 3 x 10^8 m/s. The conversion formula can be expressed as (1).

$$Distance_{\mathrm{TA}} = TA \times Propagation\ Speed \times 10^{-6} \qquad (1)$$

- $Distance_{TA}$ (meters): The $\mathrm{Distance}_{\mathrm{TA}}$ value represent the distance between the mobile station and the base station.
- TA (μs): The input value, typically measured in microseconds.
- *Propagation speed* (m/s): The speed at which radio waves travel. In free space, this is approximately 3 x 10^8 m/s.

4. Methodology

4.1) Estimating the Coverage Distance Using Antenna Characteristics

The maximum coverage distance of the base station equipped with the antenna is determined by calculating the maximum distance based on an angle referenced from half of the vertical beamwidth (*VBW/2*). *VBW/2* is the angle measured from the center of the antenna's main lobe to the point where the antenna gain drops to a specified level, typically 3 decibels. This angle represents the upper limit, ensuring that the coverage is achieved and reflects the antenna's vertical coverage capabilities.

A trigonometric method is proposed to estimate the coverage distance using the antenna characteristics outlined in Section 2. By leveraging the geometric relationships between the effective antenna height (H), and angle (θ), as visualized in Figure 2, the coverage distance can be determined.

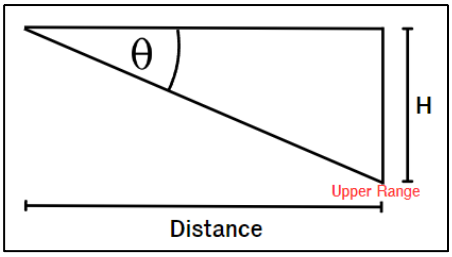

Fig.2. The relationship between distance, height (H), and θ can be described using a right-angled triangle.

The distance between a base station and a mobile station is estimated using a three-step process. Firstly, the effective antenna height (H) can be calculated by subtracting the height of the mobile station (MS) from the antenna height of the base station. This calculation can be represented by equation (2). For simplicity, we assume an MS height of 1.5 meters, which is a typical height for a human holding a mobile device.

$$H = (Antenna\ Height + DEM_{BS}) - (DEM_{Grid} + 1.5) \qquad (2)$$

The next step is to determine *angle* θ, which represents the angle used to calculate the maximum receivable distance within the upper range. This angle can be computed using equation (3):

$$\theta = (M\ Tilt + E\ Tilt) - \frac{VBW}{2} \qquad (3)$$

Finally, the coverage distance from the base station can be computed using equation (4), which incorporates the previously calculated values of H and *angle* θ.

$$Distance_{\theta} = \frac{H}{\tan\theta} \qquad (4)$$

4.2) Rate of Measured and Estimated Coverage Distance

In this study, we propose a method to detect antenna anomalies based on the comparison between measured and estimated coverage distances. Measured Timing Advance (*TA*) values-as detailed in Section 3, which are directly proportional to the distance between the base station and the mobile device, are compared to estimated distances derived from base station data (as detailed in Section 4.1). By applying Equation (5), we can identify discrepancies that may indicate antenna-related issues.

$$Rate = \frac{Distance_{TA}}{Distance_{\theta}} \tag{5}$$

4.3) Determining Acceptable Rate

The *Rate* serves as a scaling factor to estimate the actual achievable coverage relative to a reference value. The *Rate* is calculated based on a set of criteria that ensures its practical significance.

- ***Rate* < 1:** Indicates that the coverage is smaller than the reference value.
- ***Rate* > 1:** Indicates that under the given conditions, the coverage is more extensive than reference value

To establish robust criteria for *Rate* calculation, we employ distribution plots. These visual representations help identify and exclude outliers, leading to a more accurate determination of appropriate thresholds. By isolating the central tendency and dispersion of the data, we can define a range of *Rate* values that align with real-world scenarios and experimental observations

5. Results

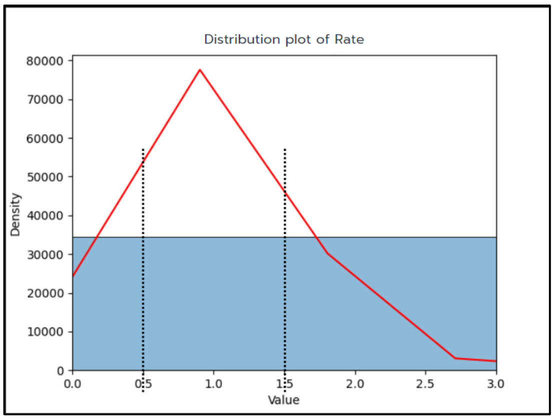

Fig.3. Distribution plot of the Rate parameter

From the distribution plot in Figure 3, we selected a range of 0.5 to 1.5 for the *Rate* parameter that the accepted coverage distance is defined. This means that the measured coverage distance is not expected to deviate by more than 50% below or 50% above the reference distance.

An anomaly detection process was applied to a dataset of 18,556 antenna cells. The analysis revealed that 4,953 cells were found to have a distance less than rate 0.5, while 4,243 cells have rate more than 1.5 These findings are visualized in Table 3.

Table 3. The number of degraded antennas

Antenna Degrade case	Number of Cell Names (cells)	Total of Cell Names (cells)
Short-Range (Rate less than 0.5)	4953	18556
Long-Range (Rate more than 1.5)	4243	18556

6. Conclusions

This study proposes a method for detecting anomalous short-range antenna performance. By comparing measured timing advance (*TA*) values with estimated distance derived from base station data, the relationship between these values using statistical distribution plots enables the identification of outlier antennas. These anomalous antennas can be targeted for maintenance, optimizing overall network performance.

7. Acknowledgement

The authors would like to thank the Advanced Info Service Public Company Limited for their generous information support, which made this research possible.

8. References

[1] Akpado K.A, Oguejiofor O.S, Ezeagwu C.O and Okolibe A.U, “Investigating the Impacts of Base Station Antenna Height, Tilt and Transmitter Power on Network Coverage,” International Journal of Engineering Science Invention Volume 2, Awka, Nigeria, 2319 – 6726, pp.32-38, July, 2013.

[2] Nikolay Dandanov, Hussein Al-Shatri, Anja Klein and Vladimir P, “Dynamic Self Optimization of the Antenna Tilt for Best

Trade-off Between Coverage and Capacity in Mobile Networks," Springer Science+Business Media, New York, USA, 92, pp.251-278, October, 2016.

[3] Raul Hernandez-Aquino, Syed Ali Raza Zaidi, Des McLernon, Mounir Ghogho and Ali Imran, "Tilt Angle Optimization in Two-tier Cellular Networks - A Stochastic Geometry approach," IEEE Transactions on Communications, Leeds, UK, 4, pp.1-16, July, 2013.

[4] Lukas Eller, Vaclav Raida, Philipp Svoboda, and Markus Rupp, "Localizing Base stations From End-User Timing Advance Measurements," Digital Object Identifier, Vienna, Austria, 1040, pp.5533-5543, January, 2022.